COST-SENSITIVE LEARNING: IN A NUTSHELL

Advanced Machine Learning

Imbalanced Learning: Cost-Sensitive Learning Part 1

- Cost-sensitive learning:
  - Classical learning: data sets are balanced, and all errors have equal costs
  - We now assume given, unequal cost
  - And try to minimize them in expectation
- Applications:
  - Medicine — Misdiagnosing as healthy vs. having a disease
  - (Extreme) Weather prediction — Incorrectly predicting that no hurricane occurs
  - Credit granting — Lending to a risky client vs. not lending to a trustworthy client.

Confusion matrix

		True class $y = 1$	True class $y = -1$
Pred. class	$\hat{y} = 1$	TP	FP
	$\hat{y} = -1$	FN	TN

Cost matrix

		True class $y = 1$	True class $y = -1$
Pred. class	$\hat{y} = 1$	$C(1,1)$	$C(1,-1)$
	$\hat{y} = -1$	$C(-1,1)$	$C(-1,-1)$

		Truth: Default	Truth: Pays Back
Pred.	Default	0	10
	Pays Back	1000	0

- In these examples, **the costs of a false negative is much higher than the costs of a false positive**.
- In some applications, the costs are **unknown** — need to be specified by experts, or be learnt.

Learning goals

- Cost matrix
- Minimum expected cost principle
- Optimal theoretical threshold

COST-MATRIX COST-SENSITIVE LEARNING: IN A NUTSHELL

- Input: cost matrix **C**
  - Classical learning: data sets are balanced, and all errors have equal costs
  - We now assume given, unequal cost
  - Goal: to minimize cost in expectation
- Applications:
  - Medicine — Misdiagnosing as healthy vs. having a disease
  - (Extreme) Weather prediction — Incorrectly predicting that no hurricane occurs
  - Credit granting — Lending to a risky client vs. not lending to a trustworthy client.

Classification y \ True Class y	1	2	...	g
1	$C(1,1)$	$C(1,2)$	...	$C(1,g)$
2	$C(2,1)$	$C(2,2)$	...	$C(2,g)$
...	...	...	...	...
g	$C(g,1)$	$C(g,2)$	...	$C(g,g)$

- $C(j,k)$ is the cost of classifying class k as j,
- 0-1-loss would simply be: $C(j,k) = 1_{[j \neq k]}$
- **C** designed by experts with domain knowledge
  1. Too low costs: not enough change in model, still costly errors
  2. Too high costs: might never predict costly classes

- In these examples, **the costs of a false negative is much higher than the costs of a false positive**.
- In some applications, the costs are specified by experts, or be learnt.

Pred. \ Truth	Default	Pays Back
Default	0	10
Pays Back	500	0

COST MATRIX FOR IMBALANCED LEARNING

- Common heuristic for imbalanced data sets:
  - $C(j,k) = \frac{n_j}{n_k}$ with $n_k \ll n_j$, misclassifying a minority class k as a majority class j
  - $C(j,k) = 1$ with $n_j \ll n_k$, misclassifying a majority class k as a minority class j
  - 0 for a correct classification

- $C(j,k)$ is the cost of classifying class k as j,
- 0-1-loss would simply be: $C(j,k) = \mathbb{1}_{[j \neq k]}$
- Imbalanced binary classification:
- C designed by experts with domain knowledge
  1. Too low costs: not enough change in model, still costly errors
  2. Too high costs: might never predict costly classes

		True class $y = 1$	$y = -1$
Pred. class	$\hat{y} = 1$	0	1
	$\hat{y} = -1$	$\frac{n_-}{n_+}$	0

- So: much higher costs for FNs

MINIMUM EXPECTED COST PRINCIPLE

- Suppose we have:
  - a cost matrix $\mathbf{C}$
  - knowledge of the true posterior $p(\cdot \mid \mathbf{x})$
- Predict class j with smallest expected costs when marginalizing over true classes:

$$\mathbb{E}_{K \sim p(\cdot \mid \mathbf{x})}(C(j, K)) = \sum_{k=1}^{g} p(k \mid \mathbf{x}) C(j, k)$$

- Imbalanced binary classification:
- If we trust we trust a probabilistic classifier, we can convert its scores to labels:

	True class	
	$y = 1$	$y = -1$
Pred. class $\hat{y} = 1$	0	1

$$h(\mathbf{x}) := \arg\min_{j=1,\dots,g} \sum_{k=1}^{g} \pi_k(\mathbf{x}) C(j, k).$$

- So: much higher costs for FNs
- Can be better to take a less probable class (Elkan et. al. 2001)

OPTIMAL THRESHOLD FOR BINARY CASE

- Optimal decisions do not change if
  - $\mathbf{C}$ is multiplied by positive constant
  - $\mathbf{C}$ is added with constant shift
- Scale and shift $\mathbf{C}$ to get simpler $\mathbf{C}'$:

	True class	
	$y = 1$	$y = -1$
Pred. class $\hat{y} = 1$	$C'(1,1)$	1
$\hat{y} = -1$	$C'(-1,1)$	0

where

- $C'(-1,1) = \frac{C(-1,1)-C(-1,-1)}{C(1,-1)-C(-1,-1)}$
- $C'(1,1) = \frac{C(1,1)-C(-1,-1)}{C(1,-1)-C(-1,-1)}$

- We predict $\mathbf{x}$ as class 1 if

$$\mathbb{E}_{K\sim p(\cdot \mid \mathbf{x})}(C'(1,K)) \leq \mathbb{E}_{K\sim p(\cdot \mid \mathbf{x})}(C'(-1,K))$$

- Can be better to take a less probable class (Elkan et. al. 2001)

OPTIMAL THRESHOLD FOR BINARY CASE / 2

- Let's unroll the expected value and use $\mathbf{C}'$:

$$p(-1 \mid \mathbf{x})C'(1,-1) + p(1 \mid \mathbf{x})C'(1,1) \leq p(-1 \mid \mathbf{x})C'(-1,-1) + p(1 \mid \mathbf{x})C'(-1,1)$$

$$\Rightarrow [1 - p(1 \mid \mathbf{x})]\,1 + p(1 \mid \mathbf{x})C'(1,1) \leq p(1 \mid \mathbf{x})C'(-1,1)$$

- Scale and shift $\mathbf{C}$ to get simpler $\mathbf{C}'$:

$$\Rightarrow p(1 \mid \mathbf{x}) \geq \frac{1}{C'(-1,1) - C'(1,1) + 1}$$

$$\Rightarrow p(1 \mid \mathbf{x}) \geq \frac{C(1,-1) - C(-1,-1)}{C(-1,1) - C(1,1) + C(1,-1) - C(-1,-1)} = c^*$$

- If even $C(1,1) = C(-1,-1) = 0$, we get:

$$p(1 \mid \mathbf{x}) \geq \frac{C(1,-1)}{C(-1,1) + C(1,-1)} = c^*$$

where

  - $C'(-1,1) = \frac{C(-1,1)}{C(1,-1) - C(-1,-1)}$
  - $C'(1,1) = \frac{C(1,1)}{C(1,-1) - C(-1,-1)}$

- Optimal threshold c^* for probabilistic classifier

- We predict $\mathbf{x}$ as class $h(\mathbf{x}) := 2 \cdot \mathbb{1}_{[\pi(\mathbf{x}) \geq c^*]} - 1$

		True class	
		$y = 1$	$y = -1$
Pred. class	$\hat{y} = 1$	$C'(1,1)$	1
	$\hat{y} = -1$	$C'(-1,1)$	0

$$\mathrm{E}_{K \sim p(\cdot \mid \mathbf{x})}(C'(1,K)) \leq \mathrm{E}_{K \sim p(\cdot \mid \mathbf{x})}(C'(-1,K))$$

OPTIMAL THRESHOLD FOR BINARY CASE

- Let's unroll the expected value and use $\mathbf{C}'$:

$$
\begin{aligned}
& p(-1 \mid \mathbf{x})C'(1,-1) + p(1 \mid \mathbf{x})C'(1,1) \leq p(-1 \mid \mathbf{x})C'(-1,-1) + p(1 \mid \mathbf{x})C'(-1,1) \\
& \Rightarrow [1 - p(1 \mid \mathbf{x})] \cdot 1 + p(1 \mid \mathbf{x})C'(1,1) \leq p(1 \mid \mathbf{x})C'(-1,1) \\
& \Rightarrow p(1 \mid \mathbf{x}) \geq \frac{1}{C'(-1,1) - C'(1,1) + 1} \\
& \Rightarrow p(1 \mid \mathbf{x}) \geq \frac{C(1,-1) - C(-1,-1)}{C(-1,1) - C(1,1) + C(1,-1) - C(-1,-1)} = c^*
\end{aligned}
$$

- If even $C(1,1) = C(-1,-1) = 0$, we get:

$$
p(1 \mid \mathbf{x}) \geq \frac{C(1,-1)}{C(-1,1) + C(1,-1)} = c^*
$$

- Optimal threshold c^* for probabilistic classifier

$$
h(\mathbf{x}) := 2 \cdot \mathbb{1}_{[\pi(\mathbf{x}) \geq c^*]} - 1
$$